

Policies and Measures in Ireland

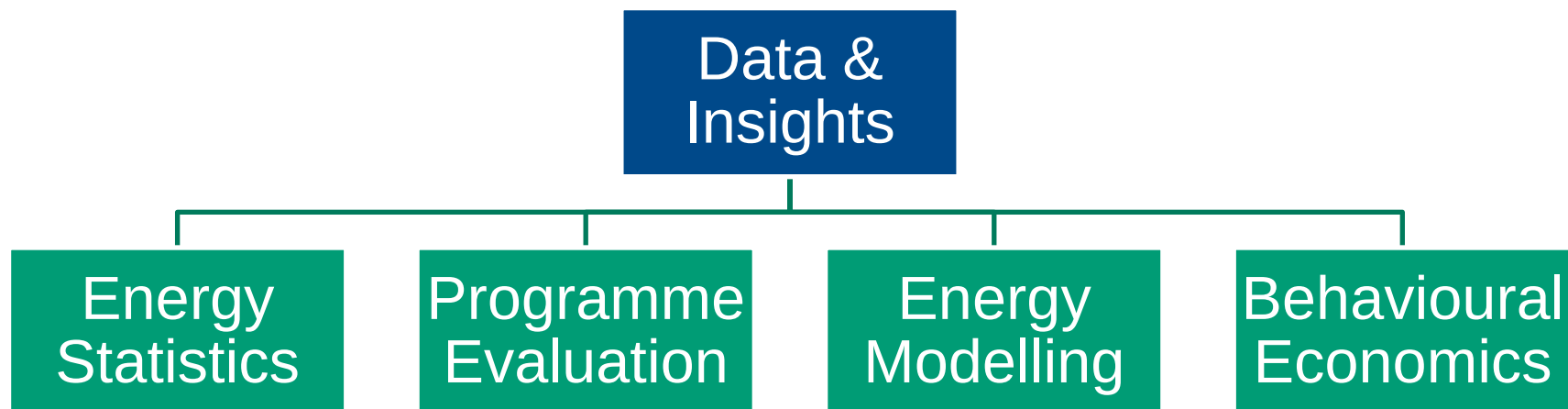
SEAI's role in providing energy-related
inputs



We work with Government, homeowners, business and communities to create a cleaner energy future

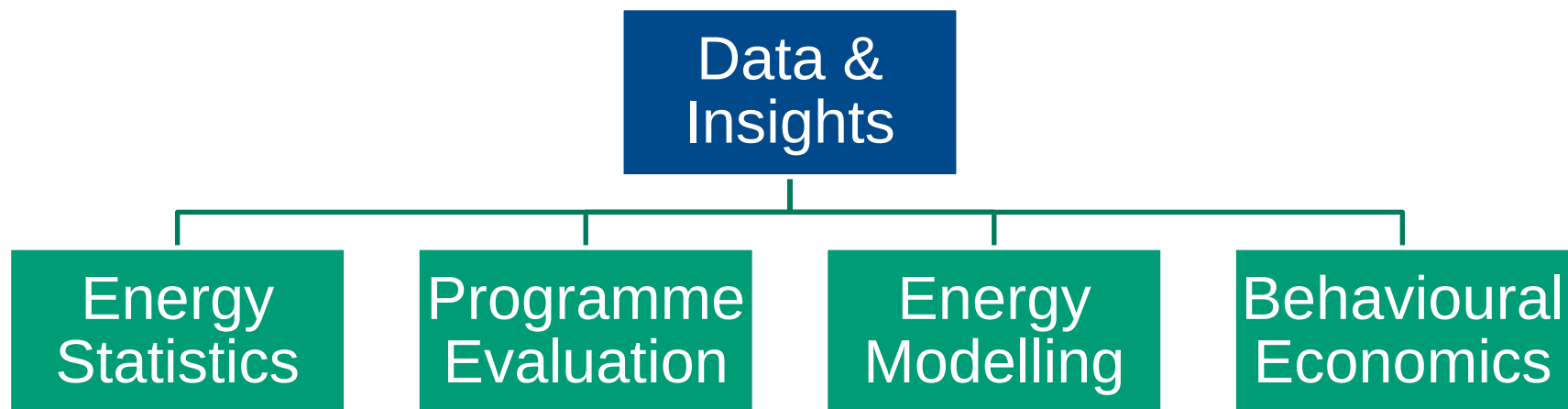


Team Functions



- Data & Insights is the analytical branch of SEAI:
 - Collect, analyse, and report national energy data.
 - Project future energy consumption.
 - Estimate policy and programme impacts.
- Ensuring decisions are guided by the best available evidence and transparent data.

Team Functions



- Two stage annual PaMs process:
 1. Estimating savings for the latest completed calendar year: Programme Evaluation Team
 2. Forecasting savings for future years: Energy Modelling Team

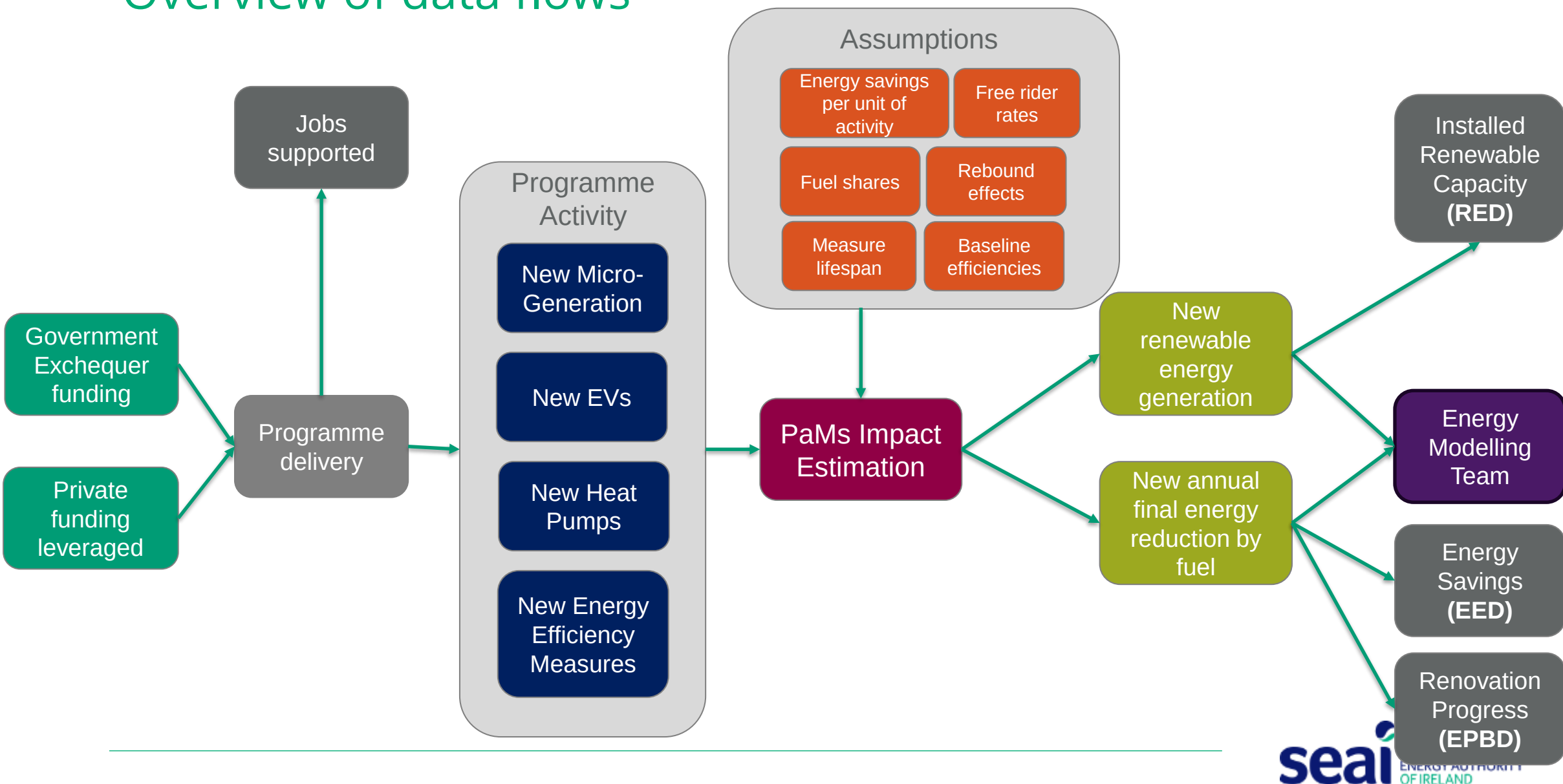
Policy and Measure (PaMs) Types

- PaMs lead to outputs in **energy efficiency improvements** and **renewable energy generation**.
- PaMs cover all energy-related sectors
 - Residential
 - Services
 - Industry
 - Transport
 - Cross-sectoral
- PaMs are delivered via:
 - Regulation
 - e.g., Energy Efficiency Obligation Scheme, Building Regulations
 - Economic incentives
 - e.g., Retrofit grants, EV grants
 - Fiscal interventions
 - e.g., Carbon tax
 - Planning
 - E.g., Transport Plan

Data Collation

- We gather data from PaMs **activity** to inform impact estimates:
 - Programme spend
 - Programme delivery, e.g., units retrofitted, renewable systems installed, etc.
- We need to make **assumptions** about certain parameters:
 - Baseline efficiencies
 - Savings per unit of activity
 - Saving per unit of spend
 - Fuel shares
 - Measure lifetimes
 - Rebound effects
- We then combine PaMs activity with associated assumptions to derive estimates of **energy savings** and **renewable energy generation**

Overview of data flows



Challenges and Continuous Improvement

Data Gaps

- Uncertainty of attribution of savings for energy networks.
- Assumed per-home savings for residential programmes.
- Estimated rebound effects.
- Delayed reporting for some programmes.

Improving Accuracy

- Incorporating real EPC improvements at the dwelling level to better reflect actual retrofit outcomes.
- Using metered energy consumption data through collaboration with other public bodies.
- Engagement with SEAI programme delivery teams to verify accuracy and timeliness of reported savings.
- Reducing manual data handling through automation in data preparation and analysis

Policies and Measures (PaMs) Projections

Energy Modelling Team
- Amy Hollands

The Energy Modelling Team

- We are a team of 10
- National Energy Modelling Framework (NEMF)
- Purposes of SEAI modelling:
 - Simulate future energy supply and demand in Ireland
 - Quantify effects of policies and measures on trends
 - Identify gaps to targets and policy priority areas
 - Fulfil national and international reporting requirements
 - Provide scenarios and risk assessment of current policy
- Core annual outputs
 - National Energy Projections (NEP)
 - Policies and Measures Projections (PaMs)

Our outputs are used by the EPA to generate the GHG emissions projections and PaMs

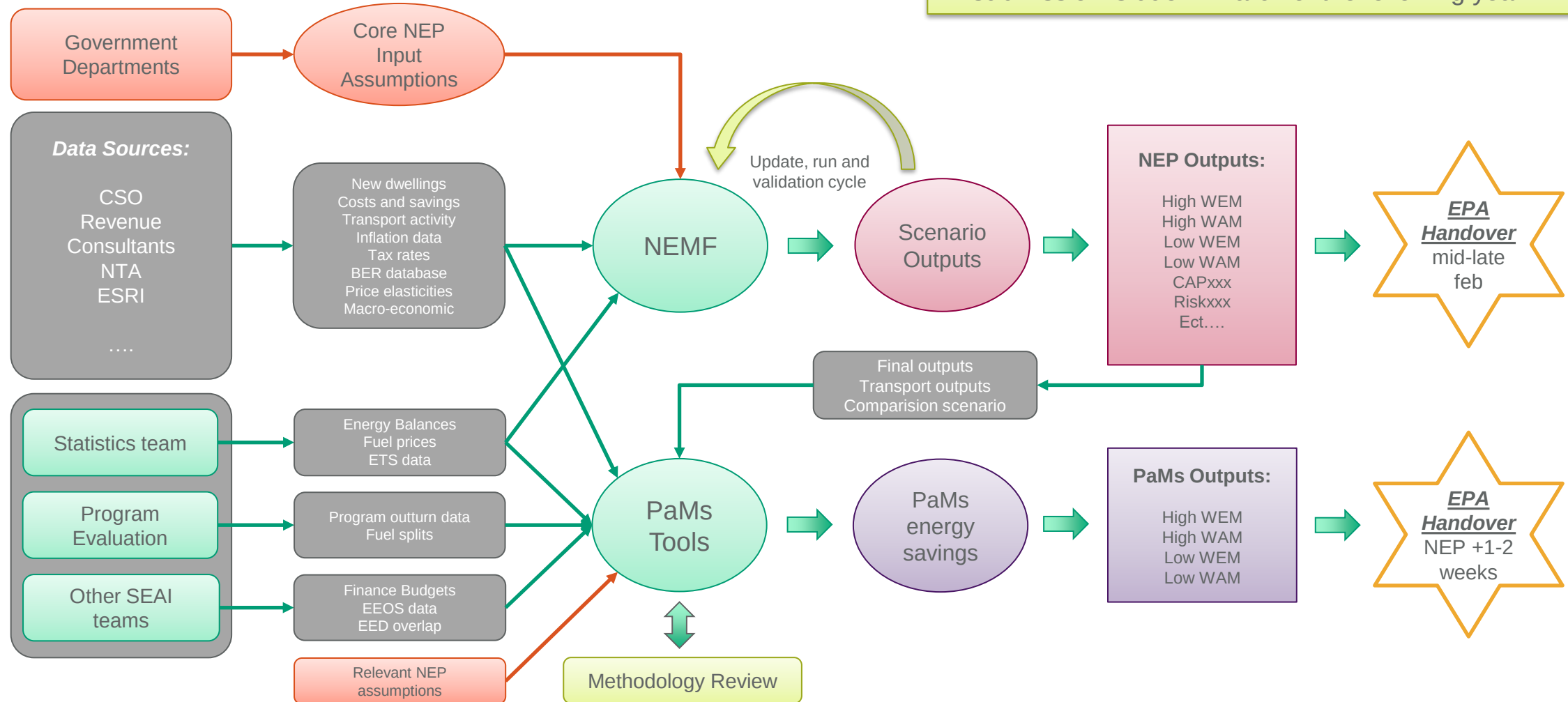


PaMs Projection Methods

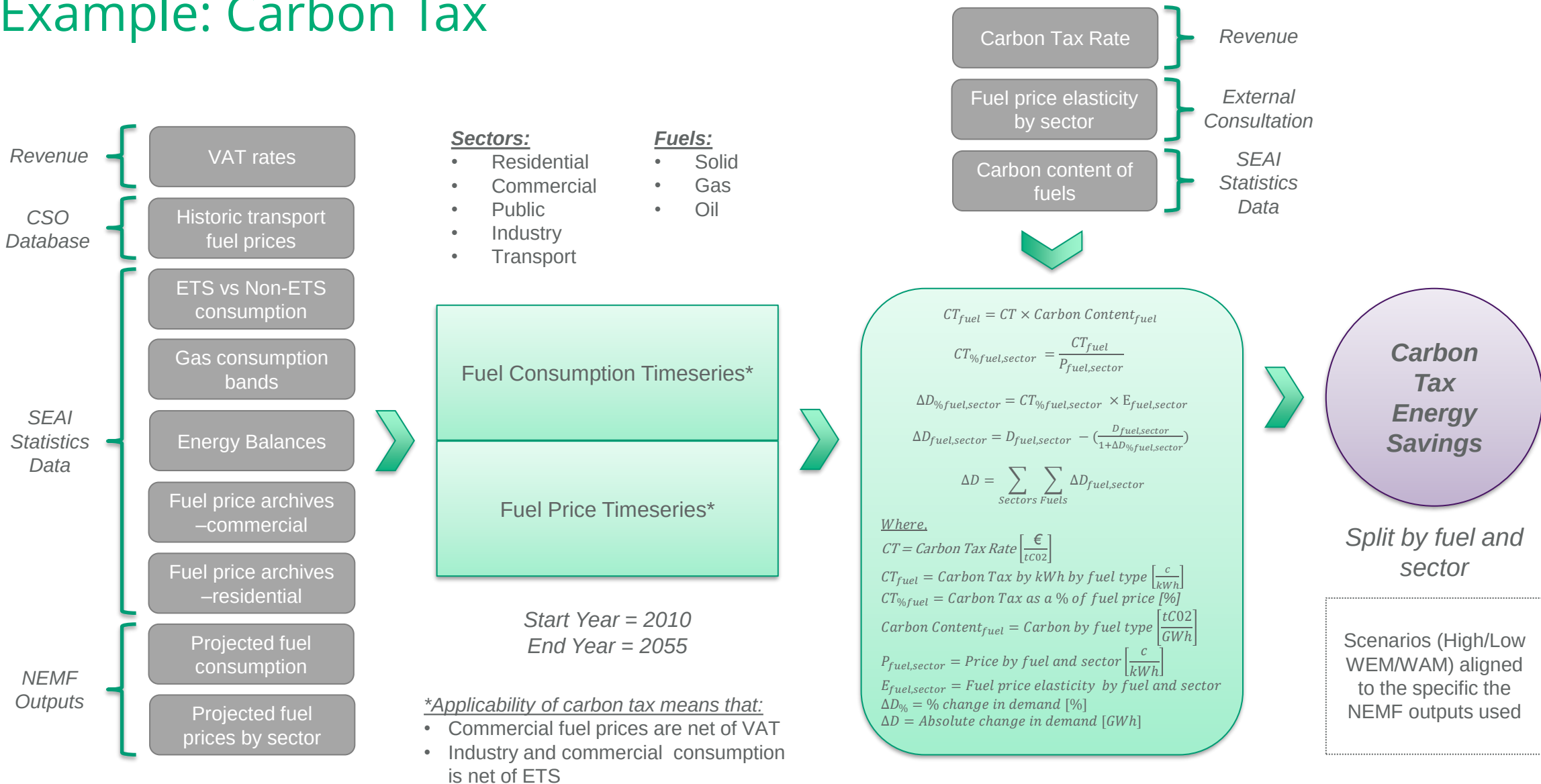
- The PaMs granularity means there is no one size fits all method.
- General objective is to compare theoretical futures – one in which the PaM doesn't exist to one in which it does.
- Aim is to capture **new annual savings** associated with a specific PaM
- Savings are reported in **cumulative terms**, counted from the inception of the PaM
- Methods include:
 - Projecting historic savings based on future budget allocation
 - Processing dedicated outputs from the NEMF projections process
 - Running comparative scenarios using the NEMF
 - Using off-NEMF modelling methods to estimate both historic and future savings
 - Combination of the above
- Methods can be reviewed and altered annually dependent on the best available information

Process flow for NEP and PaMs

Work on the projections begins in Sept/Oct. Final submission is due in March of the following year.



Example: Carbon Tax



Final Outputs for EPA handover

- Scenarios
 - High_WEM
 - High_WAM
 - Low_WAM
 - Low_WEM
- New annual savings
 - TWh/year
- Total cumulative new annual savings
 - TWh
 - ktoe
- Cumulative savings by fuel type + fuel splits
 - Oil
 - Solid fossil
 - Gas
 - Electricity
 - Biofuel
 - Ambient
 - Renewable = Biofuel + Ambient

WEM	Electricity	Gas	Oil	Solid fossil	Ambient	Biofuel
PAM	35%	40%	25%	0%	0%	0%
Public Sector Programme	35%	40%	25%	0%	0%	0%
Public Sector Capital Exemptions	30%	20%	40%	10%	0%	0%
Large Industry Programmes	35%	40%	25%	0%	0%	0%
SME Programme	85%	5%	10%	0%	0%	0%
Accelerated Capital Allowances - Services	85%	5%	10%	0%	0%	0%
Accelerated Capital Allowances - Industry	0%	40%	60%	0%	-276%	0%
CHP	-161%	474%	63%	0%	0%	0%
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Smart Meter Roll-Out - Household	0%	13%	84%	0%	0%	0%
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Smart Meter Roll-Out - Residential	-41%	27%	19%	0%	0%	0%
Supplier Obligation	54%	0%	100%	0%	0%	0%
Electric Vehicle Deployment	0%	0%	0%	0%	0%	0%
Supplier Obligation Non-Grant - Non-Residential	-45%	0%	145%	0%	0%	0%
Supplier Obligation Non-Grant - Non-Residential Transport	0%	27%	19%	0%	0%	1%
Heat Pump Supports - Domestic	0%	0%	100%	0%	0%	0%
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**Thank you
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